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Towards a New Approach for Pipeline Integrity Assessment

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Abstract

Damage detection technology is increasingly effective, e.g. through advanced vision technology. Once anomalies are detected, images and data can be analyzed in real time and remotely by experts. Pipeline integrity assessment is assisted by software platforms that are project specific. However, there is still a gap between traditional inspections in aged subsea networks, which often lack an organized database, and inspections of modern subsea systems that adopt advanced technology and rely on purpose-developed software platforms. Infrastructure's age and value, operators' preparedness, increasingly restrictive health-safety-environment requirements and economic efficiency, are all factors influencing the approach subsea operators choose to employ in each specific case, across different offshore districts, in both shallow and deep waters. The scope of the paper is to provide a link between lessons learnt and new technologies that can feed modern pipeline integrity assessment: new submarine vehicles and actuators, new sensors, big data, communication and data analytics, always in combination with advanced engineering prediction tools. The paper mainly deals with the mechanical damage caused by accidental interference, discovered during a regular survey, which may require rapid and often impacting decisions. Software platforms also support pipeline integrity assessment by maintaining a database that includes the history of each pipe joint and girth weld from design to last survey, systematically organized and regularly updated. It is now time to develop new guidelines that operators can refer to - beyond design standards (albeit influenced by them) - to take rational decisions on what-how-when to do once mechanical damage is detected, drawing on modern technology and advanced methods.

Introduction

Hundreds of thousands of kilometers of offshore pipelines are currently in service, in offshore districts that are crowded and busy. Further, new offshore pipelines cannot always avoid being routed across such congested districts, sometimes along routes affected by geo-hazards and/or severe environmental conditions. Therefore, during regular surveys, discovering anomalies, i.e. evidence of pipe damage due to external interference, is not unusual. The threats of mechanical interference on subsea are from offshore works and merchant ship traffic (Bartolini et al. 2018). As for fishing, pipelines across open sea are designed to resist fishing gear impact (DNV-RP-F111 2021). In proximity to subsea infrastructures, industrial fishing activity is commonly prohibited. Pipe diameter (small i.e. less than OD 16" vs large) and configuration (free spanning height and length vs resting on/lowered into the seabed) are relevant parameters for exposed pipelines. The thicker the pipe walls and concrete weight coating, the greater their ability to survive frequent impacts (Tura and Bruschi, 1995). Failure statistics from different international sources have been recently reviewed, to provide an estimate of failure rates for offshore pipelines to be used in risk assessment, availability analysis and contingency planning (DNV-GL TR 2017). In this paper, the subject is the sudden detection of mechanical damage on the screen of the survey operator, during a regular inspection. Damage is neither minor nor a leak or severe reduction of pipe section, i.e. medium damage requiring careful consideration: a decision on flow interruption in the short run to repair or flow prosecution that requires additional field measurements addressed to specific engineering analysis, with the aim of rationally backing any decision on future service. The latter case is the subject of the present review. Early reconstruction of geometry/shape and reconnaissance of root causes of any anomaly can be input to an early integrity assessment, from which remedial program can be initiated at the soonest. Beyond engineering judgement based on experience and competence, which are required, a quantitative assessment of the status of the asset assists survey and pipeline operators through advanced technological advances: new inspection hardware and processing software, as well as an updated basis of data from pipeline integrity management system (PIMS), that is valuable integration of the original design basis. PIMS is a software platform that includes the history of each pipe joint and girth weld, from pipe mill through installation to last survey. It assists pipeline operators to retrieve data and register new outcomes. What is missing are allowance criteria that are specifically applicable to damaged pipes, beyond design guidelines. In fact, the damage can be a permanent bending strain after pipe hooking, far beyond the design criteria, but not necessarily imposing a remedy, repair or replacement of the damaged length, for future operations. The guidelines should also account for the reduced uncertainty regarding post-design performance, after years of satisfactory service records that PIMS render available, thereby improving the operator's confidence in critical decisions.

A distinction

The treatment of the damage is different along export or transcontinental pipelines, with respect to subsea infrastructures for field development. This is reflected in the two separate recommended practices (DNV-RP-116 2021; DNV-RP-002 2021). In Figure 1, a scheme of the integrity management system is presented, reflecting the offshore Oil& Gas (O&G) industry viewpoint.

Accidental impact of dragging anchors after unintentional drop affects transcontinental pipelines. For these long strategic transcontinental pipelines, half-a-century experience is a solid ground on which modern pipeline integrity management stands. PIMS is a must, dictated by high investment costs that imply reliability over planned lifespan. This is not the same for the integrity management of subsea interfield pipe flowlines, particularly in shallow waters. The threats are mainly dropped objects and anchoring from supply and offshore maintenance vessels in proximity to bottom-fixed platforms (DNV-RP-F107 2021). Failure statistics show that failures occur mostly in proximity of the platforms, where pipe walls are thicker (not enough, burial and cover is the remedy) to meet design criteria (DNV-ST-F101 2021). In shallow water installations, even complex infrastructures, where divers are available, it looks like it is hard to employ the advanced technology developed for deep water field development. The responsibility is due to the availability of/accessibility to modern hardware and skilled operators. Sometimes, high costs might be obstacles of modernization, which can lead to the reluctance of traditional operators to change.

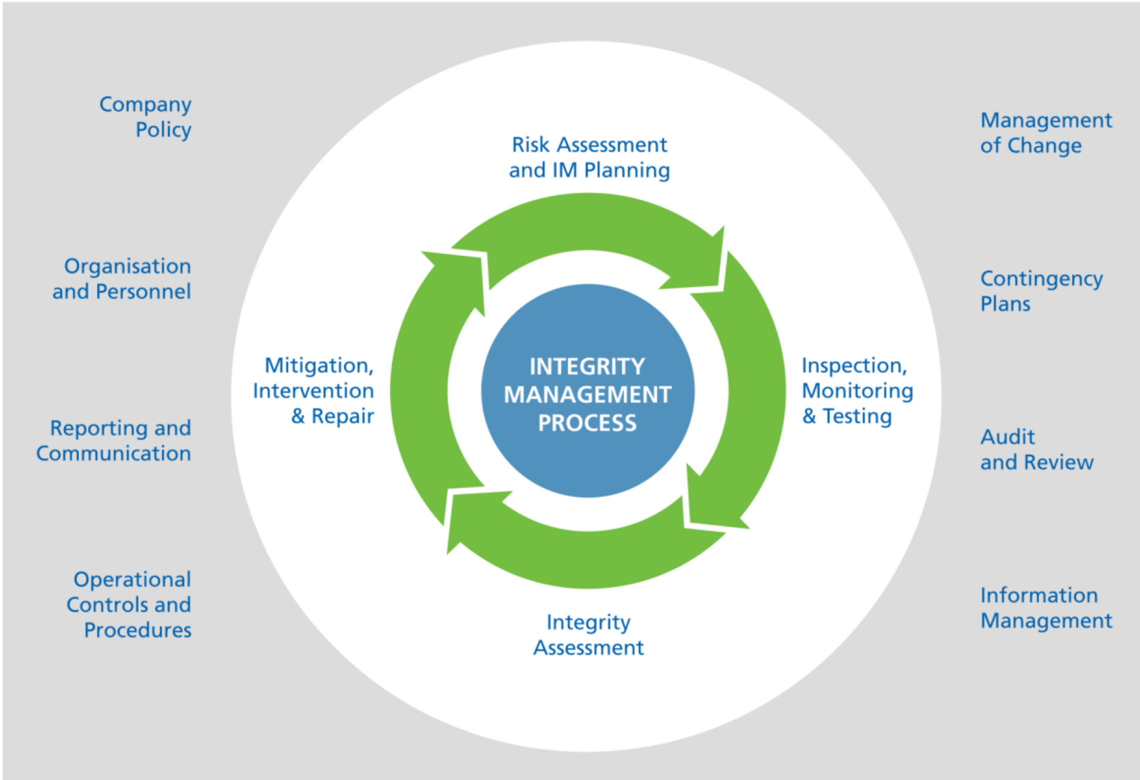

Figure 1—Integrity management of offshore assets (including pipelines) is the combined process of threat identification, risk assessment, planning, inspection, monitoring, testing, integrity assessment, mitigation, intervention and repair. It is engineering discipline, guided by recommended practice, tuned to the specific application, that rely upon experience and competence.

From detection of anomalies to damage assessment

During pipeline surveys across a brownfield, when pipe damage is detected, root causes and a measure of damage extent shall be defined at the soonest. Here, defects due to external interference are analyzed. These include gouges on the pipe wall or dents that reduce the pipe bore, sometimes permanent bending of pipe axis due to pullover or hooking after impact. Integrity of the pipeline is linked to the ability of operators to detect and assess the significance (Alexander and Brownlee 2007). An early integrity assessment is expected to define temporary remedial measures that should be exploited during the survey or to request specific tools not available during the survey, to get additional data. Later, pipeline integrity assessment is based on specific measurements and detailed engineering analyses. These outcomes are the basis for any decision on the future use of the damaged pipeline. Whether the pipeline is returned to normal use, de-classed, or, in the worst case, shut down with interruption of the flow, it is a difficult decision, often a nightmare for all operators, who are asked to ensure production, i.e. steady flow to the receiving plants. Design standards and experience drive decisions on what-to-do and how-to-do. Experience shows pipeline operators, in different offshore districts, follow different approaches to assess pipeline integrity, adopt different acceptance criteria to be met in case of pipe damage, select different methods to treat with/ repair damage. In shallow waters, outcomes have been satisfactory till now, but it is recognized that this approach is insufficient in cases of complex subsea infrastructures, because of large footprints and density of components (well-heads, flowlines, manifolds), high temperature and high-pressure (HPHT) and aggressive products. Large infrastructures imply the interconnection of different sub-systems. In deep waters and across remote offshore districts, two decades of design, installation and operation have given rise to an impressive adoption of technological advancements, purpose-built to subsea asset integrity, proven and now available for shallow-to-medium water depth infrastructures.

Case histories and lesson learnt

Failure statistics show that wall thickness (small diameter over thickness ratio, D/t, and concrete weight coating) are the main strength parameters against external interference. Bruschi and Tura (1995) have shown the correlation of failure statistics and pipe wall; Bruschi et al. (1995) have discussed the role of concrete coating as a resilient barrier against impacting objects. There are plenty of events in which minor damage and no-leaks have been detected. A diagnosis process was satisfactorily undertaken, with a final decision of minor recovery work (e.g. regarding the configuration of the pipe cover, as well as recommendations on specific controls at damaged location during future operations). Relevant documents are commonly not public. This is not the case of pipe ruptures. There have been very few events that led to the interruption of the flow (Gijerveit, Berge and Opheim 2010; Orsolato, Cherubini and Fabbri 2011; Iarla 2024) and significant repair works. Design and planning of these were accordingly performed and carried out in a short time. In the middle, there have been a lot of case histories of significant damage, but with no immediate leak (Macdonald, Cosham, Alexander et al. 2006; Omari, Loubani, Al-Mukhmari et al. 2018). Sometimes, accurate surveys and detailed engineering analysis were carried out, leading to costly and often risky repair or replacement work. But the response in time and entity have been questioned at a later stage. For pipeline operators, flow interruption is never desirable. Unfortunately, the reluctance to stop the flow was often not backed by asking for more detailed inspections, use of advanced technologies that could have offered a more realistic status of damage and provide data for advanced engineering analysis. Consequently, leaks that followed would not have occurred.

Engineering analysis

Failure mechanisms of submarine pipelines have been amply studied and tested. Statistics have shown that failure after mechanical damage due to third party interference is, along with corrosion, the main cause (Alexander and Brownlee 2007). Mechanical damage of the pipe wall, not leading to immediate leak, is commonly quoted as notch, dent or permanent deformation/curvature after excessive bending (Alexander and Brownlee 2007; Macdonald, Cosham, Alexander et al. 2006). Strength degradation of circular-resisting section is a consequence. Sometimes mechanical damage was pipe wall puncture due to sharp impacting objects or full-bore rupture due to excessive bending and stretching deformation, with consequent immediate leak (Tian, Chai, El Borgi et al. 2021; Pengchao 2025). These are cases that can be found as well documented in literature, due to the traceable consequences e.g. the mobilization of offshore vessels and repair equipment. In case of no-leak, it is difficult to get information due to reluctance to provide data that impacts reputation and economics. Data regarding shape and type of damage can be used by engineering tools to predict future development. Advanced engineering analysis can support decisions should the flow be interrupted to mitigate the effects of damage, should the damaged length be reinforced or replaced, should the severity of damage be minor, e.g. allowing for mitigation works that do not directly affect the flow. Experience shows that it would be preferable to deal with pipelines that are backed by a software platform, a database including data from design through installation and commissioning to the last survey, i.e. PIMS. It helps to support any decision with confidence when a critical condition is encountered during external surveys, and the decision might require robust justification.

Difficulties and needs

When mechanical damage is discovered, diagnostic and engineering tools are available worldwide. Obtainable data can accurately describe the geometry of the damage and feed advanced finite element modelling (FEM). These are proven engineering tools that assist decision making, through sensitivity analysis and time simulations. Testing on samples from pipes and welds at qualified laboratories, even pipe joints, may be required in case of severe damage. Material, fracture mechanics, stress analysis are involved disciplines. What are not always available are experienced and skillful resources, as integrity assessment is multidisciplinary and requires experienced (in field) and competent resources.

In Figure 2, the normal flow of investigations that follow the detection of damage until a decision, which might be twofold, is summarized: only warning during the next survey or engineering assessment based on detailed measures of damage in order to determine any additional action, assuming that no leak is present (DNV 2021). Recognizing the need for an engineering analysis, decisions must address should the damage pipeline length be replaced or should the damage be considered as not threatening future operations. A topical issue is the lack of specific guidelines that drive operators after the discovery of mechanical damage. New guidelines should be regarded: 1. reference technologies, including new advancements; 2. measurements to be performed to feed advanced engineering prediction tools (finite element models, FEM); 3. need for possible test to qualify material performance at high usage (strains). Clearly, the case of leak or full-bore pipe rupture is another story, that involves preparedness and readiness i.e. which repair technology and needed equipment to deploy, as well as which offshore vessel to transport that in the site and assist the operations (DNV 2021).

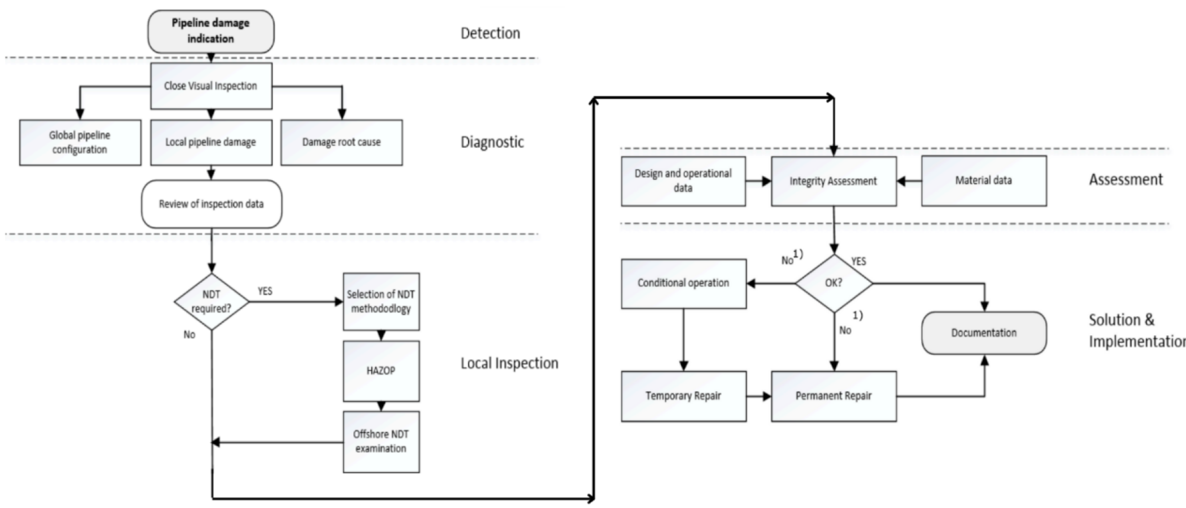

Figure 2—From damage discovery to remedial measures, based on engineering assessment. As outcome, remedies with no restriction on service: additional warning till placement of monitoring instruments; maintenance along/ across the pipeline to create protection; minor repair or reinforcement of the damaged length. Restriction or interruption of flow at the soonest are only due to leak or rupture, or in the short run due to critical condition of pressure containment. So, engineering design regards what and how to reinforce or replace the damaged length.

Mechanical damage and operational concerns

Failure statistics show that mechanical damage due to external interference is about one third of root causes of pipeline failure or misfit (DNV.GL, 2017). Records commonly regard pipe damage that led to significant maintenance work and/or pipe rupture, whether pinhole leak or full-bore (i.e. flow interruption and significant work to repair or replace the damaged length). Mechanical damage would have been far more relevant should minor damage have been included; these are recorded and suitably managed by each operator. Different classifications of mechanical damage can be found in different guidelines such as ASME B31.4 and B31.8 (Alexander and Brownlee 2007). These are commonly expressed in terms of the consequences on pipeline serviceability, e.g. ability of pigging or significant loss of flow rate due to reduced bore, and severe damage to rupture, ultimate limit state. Figure 3 shows the classification of damage commonly used for gas trunklines (Bartolini et al., 2018). At design stage, type and frequency of external interference are assessed, as well as consequent damage. In case the probability of leaks or rupture is higher than acceptance criteria, the consequences on environment and society (properties and humans) are compared with allowance criteria that are worldwide applied by industry. Should acceptance criteria not be met, protection is required. The mechanical damage depends on the route cause (e.g. the impact of a small fishing gear or the hooking of a large anchor), the pipe size (small or large diameter, concrete weight coating or anti-corrosion coating, thin or thick-walled pipe), the pipeline configuration on the seabed (free spanning or resting and partially embedded). The extent of damage depends on which pipe deformation mode is activated: either beam response mode, as in case of pipe hooking, or pipe wall deformation mode i.e. denting and/or notching of the pipe wall, as in case of local impact, or even worse i.e. a combination of both. Sometimes, concern arises from mere steel-on-steel contact between the impacting object and external surface of the steel pipe. Micro-cracks and atomic hydrogen, due to cathodic protection, might be initiators of cracks, that under cyclic load effects propagate through thickness, leading to fracture in the long run. There is an ample variety of conditions of pipe damage, difficult to embody in a simple categorization. The focus is on the reaction of operators, when pipe damage is encountered during external surveys (Amaechi, Hosie and Reda 2022). The recommendation that follows is based on the fact that the cost of engineering assessment is always less than that of potential consequences, should procedures and acceptance criteria get consensus.

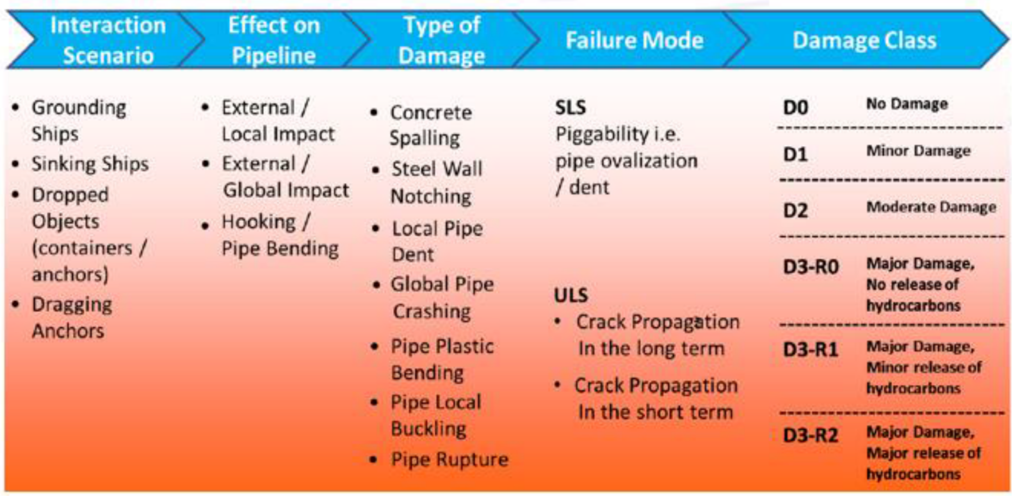

Figure 3—The classification of damage used for natural gas pipelines in Europe, that is part of the engineering analysis (frequency assessment of mechanical interference, then assessment of potential damage) carried out to define should pipeline protection be needed.

Damage to concrete weight and anti-corrosion coating

The first class of damage, less serious and frequent, is the anomaly on the concrete weight coating. It might be the spalling of small pieces of concrete found on the seabed, or simply local crushing. Both indicate that an external impact occurred. Sometimes, it might be undetected during regular surveys by operators, but video and further recognition tools applied at the office can remedy this. However, it is commonly not a major concern. Damage could be caused by a trawled fishing device, or interference from small anchors. With respect to this kind of low-energy mechanical interference (less than 5kJ for concrete coated small diameter pipes and 50kJ for large diameter-heavy wall concrete weight coated pipelines), concrete weight coating is a proven resilient barrier (Bruschi et al., 1995). Sometimes, concrete coating is requested not only as weight coating to meet on-bottom stability design, but also as protection coating from external interference in locations where mechanical interference is considered as design criteria. It is the case when a frequency greater than 10^{-4} per year kilometer is the outcome of engineering analysis of external interference risk. Nonetheless, the damage should be limited, leaving the pipe without any remedial work is a common solution (e.g., Figure 4). Unfortunately, there are interactions that can more severely damage the concrete weight coating: a significant length of pipe (one or a few pipe joints) can lose entire sectors of the concrete jacket. Damage might involve the field joint that can be torn away. The anti-corrosion coating might be either intact or lacerated, allowing seawater to contact the external steel pipe wall. In these circumstances, should the pipe not be misaligned from original settlement and pipe not be notched, damage does not necessarily lead to specific repair work of the pipe in the short run. Should similar events be frequent in one location, there might be concerns on pipe stability and protection in the long run; should the force during impact be directly withstood by the steel wall, the pipe wall might be affected; the concrete weight coating might be crushed and penetrated by metal spike or corners belonging to the impacting object, up to the pipe wall, potentially causing notching and/or denting.

Figure 4—Typical concrete damage encountered during surveys in busy (fishing and offshore works) offshore districts; understanding should pipe steel have felt the impact, possibly developing a dent, is a factor. Dedicated measurements and engineering assessments are needed.

This class of damage, considered minor, includes three sub-classes, linked to the remedial measures to be deployed:

1. no action – only increased attention during future surveys;
2. local protection cover for the relevant extent – e.g. of gravel damping or protection by typical concrete ‘mattresses’, to limit the effects of probable external interferences in the future;
3. investigation to determine should it be severe damage, with the aim of selecting the appropriate approach that might lead to remedial measures.

A library of images of concrete weight coating damage from past surveys along pipelines is available, with plenty of examples that pipeline, and survey operators can refer to; images can be even artificially built to provide a complete library of cases. Vision technology, damage recognition, artificial neural networks and machine learning can assist diagnostic decisions.

Damage to steel pipe wall

The second class of damage concerns anomalies such as smeared metal, notches, plain and kink dents observed along the affected pipe length. The steel wall is altered: steel hardening at surface and potential micro-cracking; pipe wall denting and loss of the original circular shape; smoot vs. sharp corners at dent boundaries. These damages lead to both local strength degradation and load effect intensification. In combination, there is a reduction in strength capacity. Notch and dent have been thoroughly investigated in the past, specifically for cross-country pipelines (Jones and Hopkins 1983). International guidelines try to cover the subject, providing acceptance criteria and indications on remedial measures. Extrapolation to offshore pipelines is not straightforward. Modern damage assessment for offshore pipelines is based on a detailed measurement of the dent and the use of data to feed advanced FEM. Non-linear material properties from design; advanced plasticity models allow damage analysts to calculate plastic strains that have developed. FEM can be extended to analyze how failure mode develops as a function of load effects that are expected to be sustained during the operating lifespan. Sensitivity analysis and different FEM (pipe sector, circular ring, full pipe joint) allow for a confident picture of the damaged location. This is needed to support decisions that may significantly differentiate between minor and major repair work. The footprint of external interference is detectable on the concrete weight coating, while the condition underneath the pipe wall is hard to determine. Further contribution to diagnosis might be the alignment of the pipe axis, as the impact could have shifted the pipeline out of the original settlement. Project and site-specific data, regarding the environment and human activities potentially threatening the pipeline, can be used to support decisions on how to proceed, through engineering analysis, should the detected condition need for in-line inspection. Stopping at the performed engineering stage, e.g. with a recommendation to 'freeze' the configuration and protect the exposed pipe against additional interferences, is often desirable for operators. The engineering analysis that backs such a decision shall be 'robust', additional measurements from external might be needed, leading to the necessity of removing the coating up to the steel pipe. Should such external measurements show pipe wall damage exceeding traditional allowance criteria, or advanced engineering analyses based on gathered data leave open questions on future use of the pipeline, in-line inspection is launched. Technological advancements can significantly contribute to resolving the issue. New tools provide reliable measurements of the geometry of damage. Advanced engineering analysis can be fed by high-quality data. An additional step, used by pipeline operators when the outcome is expected to be successful, is a pressure test. The effect of high pressure is the reduction of the dent, which implies improved compliance with serviceability requirements as well as mitigation of load effects at the dent. But it is not always the case. Specific engineering analyses shall be performed before moving to a system pressure test. As a summary, the four steps that are followed, for a robust assessment of integrity in case it is proposed not to repair it at the soonest, in order of increasing criticality:

1. the engineering assessment based on vision and project data;
2. the removal of coating to allow for more detailed measurements and improving the engineering assessment;
3. the system pressure test to recover part of the dent and increase the margin with respect to the allowance;
4. the internal inspection to improve the knowledge of the status of the pipe at the damaged section.

Damage/anomaly of pipe axis

A third class of pipe damage is the misalignment of the pipe axis, with or without evidence of first or second-class damage. Except in cases of macro-misalignment with visible evidence on the seabed of the original position and lateral shift, misalignment is usually detected through post-processed data. Modern technology contributes to getting an early warning: processing x-y-z data in real time might be an effective tool. It has been satisfactorily applied to assess burial depth in districts where ice gouging on the seabed is a threat (Marchionni, Bruschi and Vitali 2013). Additional considerations of possible technologies are required. As outcome, there are three classes of the misfit: 1. minor misalignment from as-built route with minor signals from seabed or pipeline; 2. major misalignment with evidence from footprint on the seabed and from minor damage on the pipe at the crest of misalignment; 3. major misalignment and strong evidence of major damage to concrete coating and pipe wall. It is known how high temperature high-pressure (HPHT) flows might be responsible of pipeline 'snaking' on the seabed, whether controlled at established locations or free, as outcome of the design (Du, Li and Wang 2023). The surveyor is prepared to measure the outcome of in-service-buckling, actions and tools to be deployed are defined. However, there are circumstances at locations where only environment or human activities are potential root causes of lateral displacements. Should a misalignment be minor, it is only recommended to pay more attention to the location during coming surveys. The presence of signals of mechanical interference on the pipe wall and the seabed can help interpretation (see fig.5). Severe signals on the pipe wall can lead to following the steps regarding the damage of the pipe wall, with the aggravating condition of plastic bending at the relevant pipe section. The engineering analysis shall account for the capacity of the pipeline to absorb plastic bending deformation, without threatening pipeline integrity in the long run. Acceptance criteria might be far beyond strain-based design in cases of major misalignments, with or without severe signals of damage to the pipe wall. They might be caused by mudslides or turbidity currents (Bruschi, 2019). When dealing with strain-based design, it is assumed that the line pipe can withstand large deformations, without threatening the pipeline integrity in the long run. Attention must be given to the residual margin with respect to loss of the bore due to sectional or local buckling i.e. serviceability limit state, at worst the loss of the bursting strength capacity i.e. ultimate limit state. Strains might be far higher than traditional or even advanced criteria from design standards allow, yet the pipeline may remain in service, without any evidence of degradation of performance. Similar concerns apply to the strains at a dented pipe. Commonly, the characterization of line pipe material at design stage does not account for working conditions at measured and simulated plastic strain. At worst, the environment might be aggressive, as in certain deep waters districts where soils are anoxic. New and specific material testing, on parent pipe and on welds, shall be required. System pressure tests can show that the residual strength of the pipe is unaltered. The competence of analysts, relying on findings from adopted technological tools and comprehensive advanced engineering analysis, must meet the stringent requirements of the company and the requests of the insurance provider. Advanced engineering analysis, off-line, based on advanced field measures, project database managed by pipeline operator (PIMS), sometimes coupled with satisfactory outcome of pressure test and laboratory testing of material performance at high usage, all contribute to back the final decision.

Figure 5—Concrete anomaly detected at the intrados of the apex of a lateral deflection. It might be caused by pull'over from the trawling board or hooking from a dragging anchor. Applied load at concrete anomaly, even should external damage be minor, might have caused pipe denting, at bending peak. Whether removing the cover to get a better indication on pipe wall deformation or performing traditional pigging, it is not straightforward. Decision asks for detailed engineering analysis.

Technology Advancements

Given the importance of subsea pipeline systems for the global energy markets and thus global economy, there has always been a strong demand for technological advancements in every aspect of subsea pipeline operation and particularly in pipeline integrity management. Similarly, constantly evolving Health-Safety-Environment (HSE) requirements also contribute to the development of subsea pipeline operation methods. Thereby, the extent of technological advancements in the domain of pipeline integrity assessment is massive. It is therefore crucial to always have a state-of-the-art overview of the available instruments to support operators. This is especially important given that such factors as the availability of hardware and skilled operators, as well as the high costs and sometimes reluctance from traditional operators, prevent the latter from employing these advancements. In the context of subsea pipelines that have been mechanically damaged, it is often the case that employing in-line inspection (ILI) methods is either impossible (due to a decreased pipe cross section, which would hinder the passage of a pipeline inspection gauge (PIG)) or strongly undesirable as it would affect production rates (Ling, Feng, Wang et al. 2023). Moreover, in certain cases, pipelines are classified as 'unpiggable' due to constraints such as small internal diameters, sharp bends, Y-junctions, or the absence of pig launchers, a situation quite common for aged subsea networks (Ho, El-Borgi, Patil et al. 2020). Therefore, in the present study, mainly external non-destructive examination (NDE) methods are touched upon.

Historical outlook

The rapid evolution of subsea pipeline maintenance and inspection technologies is essentially a pathway from employing hyperbaric diving to modern unmanned system assisted by artificial intelligence, simultaneously improving detection tools' accuracy. This evolution has also influenced the approach to pipeline system design as it went from reactive (i.e., hyperbaric diving in cases of unforeseen circumstances) to adaptive (i.e., designing tools and planning inspection activities for operation in a specific system) to interactive (i.e., designing tools and inspection activities simultaneously with the system design) (Rule and Rickey 1988). The latter approach is considered the most cost-efficient approach, given that pipeline systems have also become longer and deeper than those employed in the 1980s.

This shift towards systems deployed in deeper seas actually has been one of the factors that led to the transition from hyperbaric diving to the use of ROVs and AUVs. Throughout the 1980s, diving was the main inspection approach. Then, a certain shift had already begun to take place as the proportion of diving surveys in installation up to 200 m declined to 80% with the increased use of manned submersible systems that involved working inside an atmospheric suit at depths of up to 600 m and one atmosphere chamber surrounding subsea equipment of interest (Rule and Rickey 1988). At the time, there were also problems with inspection data quality. While the operators possessed cameras for photo and video recording, at the beginning, the tapes had been black and white (Chemisky, Menna, Nocerino et al. 2021). Much greater quality subsea photos and videos became available with the introduction of underwater color video in 1980 (Faulds 1982). All of these further improved diver-to-surface and ROV-to-surface communication (Faulds 1982). Given that the manned submersible systems already had manipulators to be guided by operators inside, closer to the end of 1980s, ROVs became commercially available and also equipped with manipulators to enable performance of certain tasks, such as debris removal lift lines' attachment, and even Sidescan Sonar Surveys (Rule and Rickey 1988; Shand 2019).

There has been also a shift in used detection methods. One example is the transition from radiographic NDEs for pipeline weld inspection activities, which were a dominant approach in the 1980s, to using UT techniques since radiographic testing did not allow to measure the defect height and involved significant safety concerns (Moles 2004). At the same time, however, UT testing has been developing as well, since initial methods were not reliable, which led to the introduction of Time-Of-Flight-Detection (TOFD) methods (Moles 2004). Moreover, early amplitude-based ultrasound testing often required high operator skill and detected mainly the shallow defects (Faulds 1982). The same problem of skill requirement was also relevant for magnetic particle inspection (MPI) that was common in the 1980s (Watt, Walther and Walther 1989). This led to extensive research efforts on increasing accuracy of NDE techniques in the period between 1980s and early 2000s, which led to automated UT becoming more dominant, while also resulting in greater accuracy of existing NDE methods (Moles, 2004). Thereby, by the 1990s, oil and gas operators were employing a variety of electromagnetic inspection methods, including MPI, Eddy current (EC) testing and ACFM (Dey, Ogunlana and Naksuksakul 2004).

In the 1990s, the shift towards greater employment of ROVs continued, and ROV-equipped instrumentation involved a considerable variety of instruments such as low-light and twin boom-mounted color video cameras, dual-headed scanning sonar, an obstacle avoidance sonar and an electro-magnetic pipetracker system (Valdez, Bayazitoglu, Weiss et al. 1997). Moreover, closed-circuit television cameras have also become common for ROV inspections, which further contributed to the ability to make real-time decisions. This shift was partially motivated by safety considerations and high costs of employing divers for inspection activities, and partially by greater depth at which the subsea pipes were located (Dey, Ogunlana and Naksuksakul 2004). It was an important factor in gradual decrease in the use of MPI techniques since early 1990s as it is limited to accessible ferromagnetic surfaces and generally requires direct contact (Nordbo 1986). At the same time, in the period between the late 1990s and 2010s, fiber optic sensing became a popular method for pipeline inspection, including leakage detection purposes (Peng, Jiang, Wen et al. 2020). Fiber optic sensing is an advanced monitoring technology increasingly used in subsea pipeline integrity management to provide continuous, real-time information on the operating and structural condition of underwater pipelines (Peng, Jiang, Wen et al. 2020; Sharma, Singh, Gupta et al. 2021). By deploying optical fibers along the pipeline — either incorporated during construction or retrofitted to existing infrastructure — it is possible to obtain measurements of parameters such as strain, temperature, vibration, and, in some cases, pressure, without the need for electrical components in the subsea environment. The principle of operation relies on transmitting light through the fiber and analyzing changes in its properties, using onshore or topside interrogation units. Their increased use starting from the late 1990s is due to their immunity to electromagnetic interference and high sensitivity for dynamic strain measurements (Peng, Jiang, Wen et al. 2020). Their capacity for real-time event detection supports early intervention, enhances operational safety, and contributes to the extension of pipeline service life through informed maintenance planning.

Various configurations of fiber optic sensing systems are available. Fiber Bragg Grating (FBG) arrays offer discrete, high-precision measurements at predetermined points, while distributed sensing technologies —such as Distributed Acoustic Sensing (DAS), Distributed Temperature Sensing (DTS), and Distributed Strain Sensing (DSS)—allow the entire length of fiber to function as a continuous sensor throughout the pipeline (Sharma, Singh, Gupta et al. 2021). These techniques are capable of detecting a range of integrity threats, including leaks, anchor strikes, fishing gear interaction, seabed instability, and deformation caused by internal or external loading.

In the same period, the general trend of employing unmanned inspection activities and thus more frequent use of ROV/AUVs due to safety and economic reasons of operating in deeper seas continued, while the shortcomings of previous technologies (e.g., low quality/precision data) was being resolved (Zhang, Han, Han et al. 2024). The novelty, however, was the increased necessity of real time data communication, which became available primarily due to the usage of fiber optic sensing technologies that could be monitoring the whole span of subsea pipelines in real time (Sharma, Singh, Gupta et al. 2021). In the context of pipeline mechanical damage, this innovation is of critical importance as it further assists the operators save time and resources by enabling swift response to any anomaly.

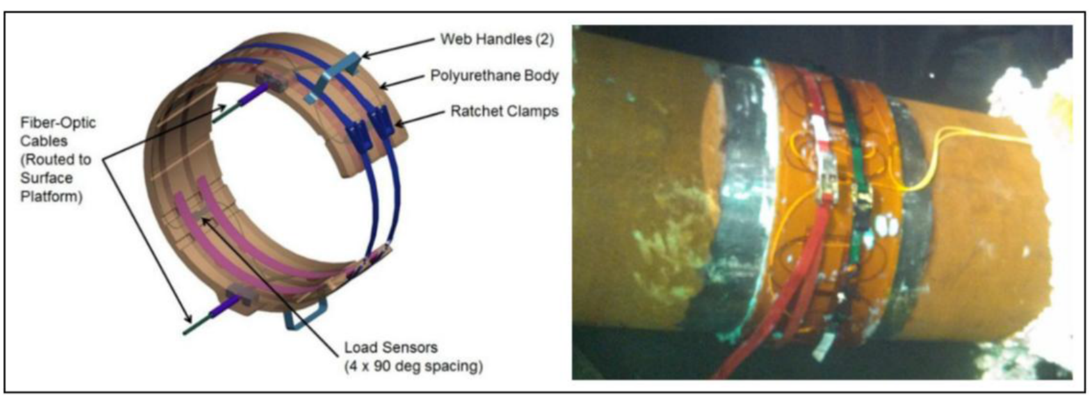

Figure 6—FBG system layout (Ho, El-Borgi, Patil et al. 2020)

State-of-the-art

The two trends (shift towards ROV/AUV inspection activities and real-time data communication) mentioned in the last chapter are still being observed as, in the past decade, there have been important advancements in the domain of subsea pipeline ROV-/AUV- and ML-assisted NDE. That is, while some inspection activities are still carried out by divers up to a specific depth (depending on local regulations), the usual methods involve the use of ROVs and real-time sensors (Sharma, Singh, Gupta et al. 2021). These ROVs are further equipped with tools specific to certain survey objectives. That is, in the context of visual inspections, as per DNV-RP-F116 (DNV 2021), the GVI, DVI/GVI XTD and CVI are further accompanied by High Precision Survey (HPS), which have to determine the absolute position and relative movement of the pipelines. This gradation of surveys highlights the diversity of inspection tools and precisions available nowadays. This is particularly relevant to currently available ROVs, which range from small Observation Class units, primarily suited for visual inspection, to larger Work Class systems capable of accommodating and operating a wide array of heavier, more sophisticated sensors (Chemisky, Menna, Nocerino et al. 2021).

In the context of visual inspection, the advancements currently employed involve laser scanning and close-range photogrammetry for accurate mapping, localization and reconnaissance (Chemisky, Menna, Nocerino et al. 2021). Ensuring proper and accurate data gathering during subsea inspection remains challenging due to factors such as the proximity between vehicle and target, the narrow camera field of view available to ROV operators, variable visibility conditions, scene homogeneity, and the absence of distinctive reference points (Chemisky, Menna, Nocerino et al. 2021). In this regard, the post-processing of previously mentioned laser scanning and photogrammetry data can be time-consuming, while in many cases delaying quality assessment is operationally impractical (especially in cases of mechanical damage, which may require swift reaction) (Menna, Torresani, Nocerino et al. 2019). Therefore, Simultaneous Localization and Mapping (SLAM) technologies are forming essential components of near-term and mid-term inspection strategies (Menna, Torresani, Nocerino et al. 2019). These approaches not only support the broader shift toward autonomous inspection systems but also ensure the creation of precise and reliable digital archives that shall be of incredible use should mechanical damage be detected during a regular survey.

Moreover, in recent years, despite requiring post-processing, integrated photogrammetry systems for subsea applications have been developed for deployment on ROVs or by divers, enabling high-precision 3D reconstructions. These systems may rely solely on photogrammetry or combine data from complementary sensors. Currently used photogrammetry systems have achieved millimeter-scale accuracy without physical seabed markers through a trifocal camera system capturing synchronized high- and low-resolution imagery (Chemisky, Menna, Nocerino et al. 2021). Some of the advanced solutions currently allow stereo baseline adjustment to reduce triangulation error and improve overlap for close-range targets (Chemisky, Menna, Nocerino et al. 2021).

The employment of ROVs for pipeline surveying enables an increase in data quality as ROVs, in contrast to divers, may be designed and thus purpose-developed to accommodate specific types of inspection tools that previously were challenging due to skill and/or safety concerns. One of the prominent examples includes more frequent use of radiographic methods, which were once deemed more dangerous and less accurate than acoustic methods (Ho, El-Borgi, Patil et al. 2020). To be more precise, radiography is again widely applied in pipeline inspection for the detection of internal defects such as cracks, corrosion, and weld imperfections through imaging of wall-thickness variations and post-processing via digital radiography (Ho, El-Borgi, Patil et al. 2020).

Another advancement currently employed is the use of pipeline crawlers (Ho, El-Borgi, Patil et al. 2020). Similar to smart PIGs, these crawlers can be equipped with a variety of inspection instruments to assess pipeline integrity (Ho, El-Borgi, Patil et al. 2020). Unlike pigs—which are propelled internally by a hydraulic pressure differential—external crawlers require alternative propulsion methods, such as onboard electric motors, towing by an ROV, or manual positioning by a diver (Ho, El-Borgi, Patil et al. 2020). At the same time, however, their operational envelope may include traversing vertical sections such as risers (Ho, El-Borgi, Patil et al. 2020). Commercial setups include subsea inspection tools that use electromagnetic acoustic transducers (EMATs) to perform 360° volumetric assessments of pipelines, flowlines, risers, and other subsea assets via ROV, delivering real-time wall condition data in a single deployment without halting production (Zhang, Zheng, An et al 2019).

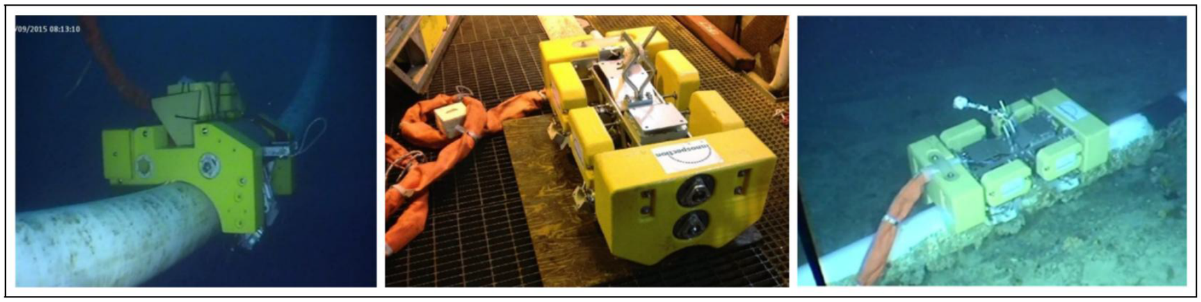

Figure 7—Crawlers inspecting subsea pipelines (Ho, El-Borgi, Patil et al. 2020)

Additionally, for quite some time already, AUVs are gaining more presence in the industry with the aim of improving efficiency and optimizing inspection costs (Chemisky, Menna, Nocerino et al. 2021). AUVs eliminate the need for continuous tethering to a support vessel, which can substantially reduce operational costs associated with vessel time and crew hours compared to ROV-based inspections, thereby further motivating relevant research activities (Ho, El-Borgi, Patil et al. 2020).

Recent developments have expanded AUV roles into tasks traditionally handled by divers or ROVs, including corrosion assessment and detailed visual surveys. Ongoing research focuses on long-duration subsea residency with today's AUVs being designed to remain docked at a subsea station for extended periods (Ho, El-Borgi, Patil et al. 2020). Such docking stations not only recharge the vehicle but also offload collected data, enabling near-continuous inspection availability and data communication (Ho, El-Borgi, Patil et al. 2020).

Lastly, the technological advancements of today already include ROVs being able to conduct minor repair work. In recent years, advances in robotic manipulation, vision systems, and power delivery have made ROV-assisted underwater hyperbaric welding more feasible (Nauert and Kampmann 2023). While they are not ubiquitous in the industry yet, the general trend has already been established with enormous research efforts being dedicated to employing AI for wet welding operations by ROVs and AUVs (Nauert and Kampmann 2023). These efforts have already yielded subsea robots capable of laser and friction welding (Nauert and Kampmann 2023).

Future outlook

Although available already today, the future of subsea pipeline survey activities seems to be further shifting towards detection methods' synergy to combine advantages and tackle the shortcomings of each individual method. That is, while some of them can be overcome by integrating technological advancements through hardware improvements, some are being tackled by multimodal fusion, which is expected to achieve much greater integration in the industry.

Without a doubt, some advancements that have already been presented and yielded accurate results did not employ hybrid solutions but were rather breakthroughs in themselves such as the subsea Computed Tomography (CT) scanner capable of performing online tomographic reconstruction of pipelines at depths up to 3000 m (Zhang, Zheng, An et al 2019). These still niche solutions may also change the industry's perspective on subsea pipeline inspection and may actually be of great use in cases of mechanical damage. Nevertheless, given that they typically represent outliers and thus "out-of-the-box" thinking, it seems that the subsea pipeline inspection future is primarily in combining the strengths of traditional methods.

Moreover, pipeline integrity assessment increasingly relies on advanced inspection tools and data analytics, requiring the acquisition, processing, and integration of large, complex datasets from multiple sources over extensive pipeline networks. In this regard, technological advancements in data management and particularly in ML-assisted data denoising and pattern recognition are already shaping the future of subsea pipeline integrity assessment.

Hybrid solutions. Multimodal data fusion has become a valuable strategy for integrating information from diverse sensor modalities, enabling more accurate and comprehensive measurements across various applications, including subsea inspections (Zhang, Han, Han et al. 2024).

This method has been commonly used for combining acoustic and optical sensors. By leveraging their complementary advantages, optical sensors capture high resolution, color rich images, while acoustic sensors provide accurate range measurements even in low visibility conditions (Zhang, Han, Han et al. 2024). This represents a significant advancement in addressing the challenging imaging conditions commonly encountered in subsea environments, such as scattering, attenuation, limited visibility, and low-light conditions. Additionally, image acquisition by underwater platforms and divers often introduces motion-related distortions—such as blurring and jitter—that further compromise image quality. A key optical technique in this field is stereo vision, which estimates depth by comparing two images taken from slightly different viewpoints (Zhang, Han, Han et al. 2024). By detecting and matching corresponding features in both images, the system calculates object distances through triangulation. One widely used stereo vision method is the Semi Global Block Matching (SGBM) algorithm, which efficiently computes depth maps by aggregating matching costs across multiple paths (Zhang, Han, Han et al. 2024). Recent research by Zhang, Han, Han et al. (2024) enhanced underwater stereo vision by integrating sonar data directly into the SGBM cost computation and aggregation steps. This hybrid optical–acoustic method significantly improved robustness in extreme underwater conditions, reducing measurement errors to as low as 1.6% in experiments.

Another example is the usage of dry cabins in combination with structured light scanning (SLS) proposed by Zhu, Lin, Jin et al. (2025). The dry cabin is formed by expelling seawater using high-pressure gas and pumps, enabling rapid scanning during slack-water conditions. Applied successfully to expandable subsea pipelines in China, the method involves the use of structured-light 3D scanning that delivers high-precision, editable datasets, and versatile processing options.

Similarly, this hybridization shall take into account the possibility of transmitting data between inspection entities. For example, the integration of AUVs with underwater acoustic communication networks offers a pathway toward more advanced cyber-physical monitoring systems (Ho, El-Borgi, Patil et al. 2020). In this envisioned setup, fixed subsea sensor nodes continuously gather pipeline condition data and transmit it acoustically to passing AUVs, which in turn consolidate and relay the information (Ho, El-Borgi, Patil et al. 2020). This architecture could greatly enhance real-time situational awareness for subsea asset integrity management.

This trend of hybridization also involves hybrid subsea vehicles (including, resident or Seabed-Hosted Vehicles (SHVs)), which are gaining significant attention from operators due to their potential to reduce dependence on costly surface support vessels (Chemisky, Menna, Nocerino et al. 2021). Stationed at a dedicated subsea base positioned near the infrastructure, SHVs can perform a wide range of IMR activities within their operational range. These systems are versatile, capable of carrying their own energy supply, operating in conventional supervised ROV mode, or executing fully autonomous missions for specific tasks. Several commercial SHVs already can remain deployed for up to 12 months with a 10 km operational radius and in depths reaching 6,000 m (Chemisky, Menna, Nocerino et al. 2021).

Big data and AI incorporation. Pipeline operators increasingly rely on integrated systems capable of aggregating all available information over the full life cycle of a field (Chemisky, Menna, Nocerino et al. 2021). Thus, with the rise of technological capacity in the areas of data analysis and inspection tools, the pipeline integrity assessment operations consist of complex procedures of data acquisition, management and utilization for eventual diagnosis of pipeline fitness for purpose. Moreover, given that pipeline integrity management operations involve collection of data from a variety of inspection tools (e.g., ILI, NDE) for hundreds or thousands of kilometers, there is always a necessity for proper management of large volumes of complex data being generated at high rates (Rachman, Zhang and Ratnayake 2021). In addition, harsh subsea environments inevitably lead many sensors employed during inspection and monitoring activities to generate noisy data that requires further processing (Spahic, Poola, Hepso et al. 2023).

In the context of signal denoising, various methods have been employed in pipeline management systems. One of the common techniques involves the usage of the wavelet transform (WT), which is a signal processing method that analyzes data in both the time and frequency domains simultaneously by decomposing a signal into scaled and shifted versions of a mother wavelet (Ergen 2012). This makes it especially useful for detecting mechanical damage in subsea pipelines, where events such as impacts, denting, crack initiation, or joint failure often generate short-lived and irregular signal patterns (Zhang, Zheng, An et al. 2019). For example, in guided wave ultrasonics (GWT), wavelet analysis enhances defect echoes from dents, weld misalignments, or structural discontinuities along the pipe wall. When integrated with distributed acoustic sensing (DAS) or strain-based monitoring systems, WT can help separate genuine mechanical stress signatures from environmental noise, improving detection accuracy.

Similarly, noisy data may also interfere with real-time decision-making processes employed by autonomous underwater systems used for inspection (and maintenance) (Spahic, Poola, Hepso et al. 2023). Given gradual yet inevitable rise of AUVs, pattern recognition and particularly employing algorithms in subsea pipeline inspection is an area that certainly will shape the future of pipeline integrity assessment. Similarly, pattern recognition shall play an important role in the evolution of optical fiber sensors. Peng, Jiang, Wen et al. (2020) employed deep neural networks particularly for this aim in their study on improving the accuracy of mechanical damage detection by DAS systems that rely on detecting Rayleigh backscattering signals—weak reflections of light caused by microscopic variations in the fiber optic cable's glass structure (Ho, El-Borgi, Patil et al. 2020). Given the diversity and volume of DAS data, an adaptable analytics tool is required for effective event identification and feature classification (Peng, Jiang, Wen et al. 2020). For instance, the acoustic signature of infrastructure operating in the vicinity of the pipe is distinguishable from the high-pitched noises characteristic of mechanical damage to a pipeline by an anchor. Machine learning approaches offer a promising solution by managing large volumes of DAS data and uncovering patterns related to both external threats and internal pipeline corrosion or erosion, enhancing the reliability and specificity of pipeline monitoring systems.

Further, the software used for pipeline integrity management shall also employ proper data compression techniques in order to avoid overloading the systems designed for feature extraction and pattern recognition. One of the most effective strategies employed is the Kernel Principal Component Analysis (KPCA), which maps data into a high-dimensional feature space (via kernels) and performs linear dimensionality reduction, thereby optimizing data compression in the context of subsea pipeline inspection data that is not linearly separable (Zhang, Zheng, An et al 2019). Similarly to data denoising, machine learning techniques are being massively employed for data compression mainly through feature selection via genetic algorithms and fully convolutional neural networks (Huang, Chen, Li et al. 2024).

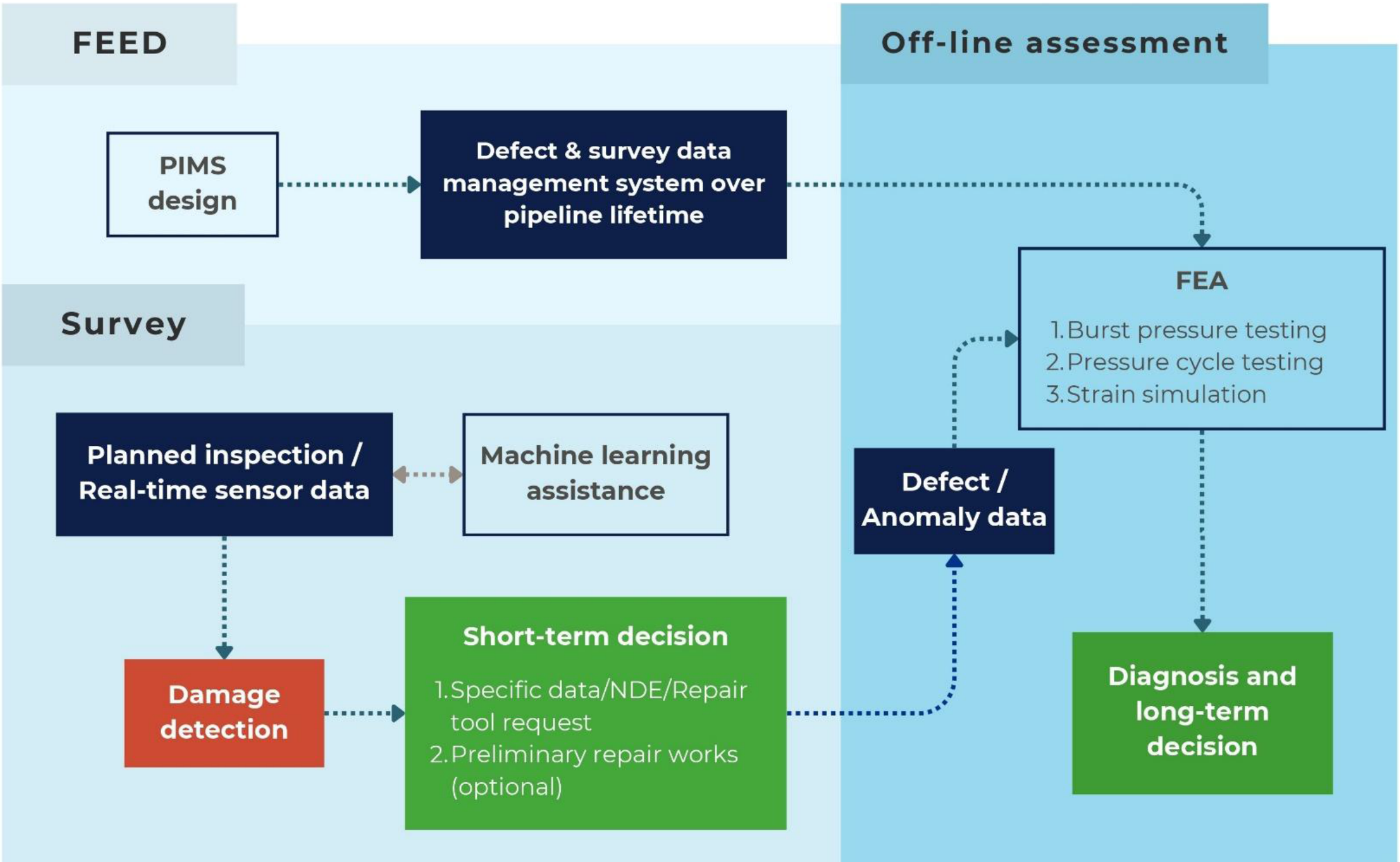

Figure 8—Proposed methodology for pipeline integrity assessment

Conclusions

There are four decades of experience with software platforms dedicated to pipeline integrity management, PIMS. They mainly regard strategic natural gas pipelines, transcontinental. Preparedness to tackle accidental events is a factor. Software platforms for integrity management of inter-field pipelines are more recent; they date back no more than two decades. Motivation has been life extension of infrastructures in service, further developments of traditional offshore fields, as well as the operation of new networks of inter-field and export pipelines in deep and ultra-deep waters, across remote and harsh offshore districts.

Technology

Offshore pipelines are long pressure vessels, currently in order of one million kilometers-years in service. A reference for assessing the quality of pipeline infrastructures, whether traditional or in ultra-deep waters, might be the analysis of failure statistics. Ruptures of pipelines remain a typical low-frequency and high-consequence incident. Inspection, maintenance and repair are recognized to contribute to ensure operation over the planned lifespan. Asset integrity and functional performance are well-structured, strictly tuned to the different uses and different ages of pipelines. Relevant differences are linked to length, environmental districts, transported flows, transport parameters, installation methods, control technologies, design solutions, line pipe material and fabrication technology. The main outcome of failure statistics is that offshore pipelines have a significant tolerance to external interference and mechanical damage. There are pipelines that are notched or /and dented, sometimes permanently bent and plastically deformed, that are satisfactorily in service. Offshore pipelines are commonly thicker than onshore pipelines. Pipe wall thickness and line pipe material are principal factors of safety against external damage. Concrete weight coating, commonly designed to provide on-bottom stability, significantly contributes to withstanding minor interferences. It is enough in most circumstances, e.g. where fishing activity might be severe and frequent; it is not enough to survive the severe mechanical interference from dragged anchors, accidentally dropped from merchant ships, as evidenced by recent accidents.

Detection - measurement – analysis

During regular surveys, discovering anomalies, i.e. evidence of pipe damage due to external interference, is not unusual. In case of pipeline damage from external interference, lessons learned have shown that the engineering analysis of the damage is a must. There are damages that are minor, not requiring immediate remedial measures, and damages that lead e.g. to replacement of pipeline lengths. In the middle, there are damages for which any decision regarding mitigation measures requires a specific assessment. The outcome determines how the present and future use of the pipeline should be impacted. All engineering steps of pipeline integrity assessment are crucial, from measurements regarding the shape and the nature of damage to the engineering analysis performed to simulate the working conditions, in the short and long run. In certain circumstances, laboratory tests might be requested to understand the actual performance of the steel, after significant plasticity, commonly not specifically addressed at design stage. Advanced engineering is less expensive than undue flow interruption, which is the nightmare for all pipeline operators. Confidence in future operations requires the best engineering effort. Inspection tools are a factor. Modern sensors for internal and external inspection are the outcome of impressive advancements, driven by/linked to digital transformation. Improved capacity of data gathering, storage and communication can significantly contribute to pipeline integrity assessment, e.g. by providing data to accurately describe the geometry of the damage that can be used by advanced finite element modelling.

Advances

Nowadays, digital twin is a ‘mantra’ of all new offshore installations. Except for intrusion, when there are conditions that require continuous monitoring, now real time inspection for offshore pipelines does not appear to be a real necessity. But there are cases for which uncertainties at design stage may lead to costly choices, that are often not resolutive in the long run. Nonetheless, the engineering analysis of the damage is a factor. Advanced finite element models can be used to simulate the damage and to define the residual state of stress and strain that follow. As input, the geometry and mechanical performance of steel. Damages are different and similar, at the same time. Advanced calculus helps to define the performance of the pipeline in residual life, should it be within allowance criteria, with or without remedial measures. Remedial measures are intended as indirect work regarding the surrounding medium, e.g. the configuration of the pipeline in proximity to damage is ‘frozen’ to avoid any worsening of the damage under future operating loads. Any work on pipe walls might be of impact for normal operations; if any, careful engineering assessment is requested. When there is evidence of severe plastic deformation, far beyond the one that can be envisaged at design stage, it is needed to understand the future performance of line pipe material and welds in the plastic domain, outside the standard characterization of material performance. This is a case that moves the operator to the laboratory testing on samples of available line pipe, as well as of new welds purposely performed on available pipe joints by adopting the same equipment used during installation, as well as the adopted qualified procedures. It can be also requested to reproduce damage like the actual, with the aim to perform full scale strength tests.

Needs

The complexity of the three steps moves operators to ask whether guidelines should be in force, as purpose developed. For offshore pipelines, it is commonly preferred to perform a specific engineering analysis and, when and if needed, specific material performance characterization. Design guidelines can be used only as a reference, to be compared with the outcomes of advanced engineering analyses of damage. Operating records of modern pipelines, the ability to provide a realistic picture of precedent life for old pipelines, and the outcome of specific engineering assessment shall be rationally used together as a reference for future operations. Therefore, it is remarked the lack of specific guidelines that drive operators after the discovery of mechanical damage. New guidelines should regard the use of new technologies, including new measurement tools to be performed, as well as advanced engineering prediction tools (finite element models, FEM) that shall be used, and, if possible, tests to qualify material performance at high usage (strains). The available basis for decisions can be known as far more valuable than the original design basis. There is room for developing new guidelines for the integrity assessment of offshore pipelines, creating a relational framework between new technology and engineering prediction tools.

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